

Администрирование локальных сетей

Лабораторная работа №13

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Цель

Выполнение организационно-подготовительных мероприятий по обеспечению взаимодействия через сеть провайдера посредством статической маршрутизации локальной сети с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

Выполнение лабораторной работы

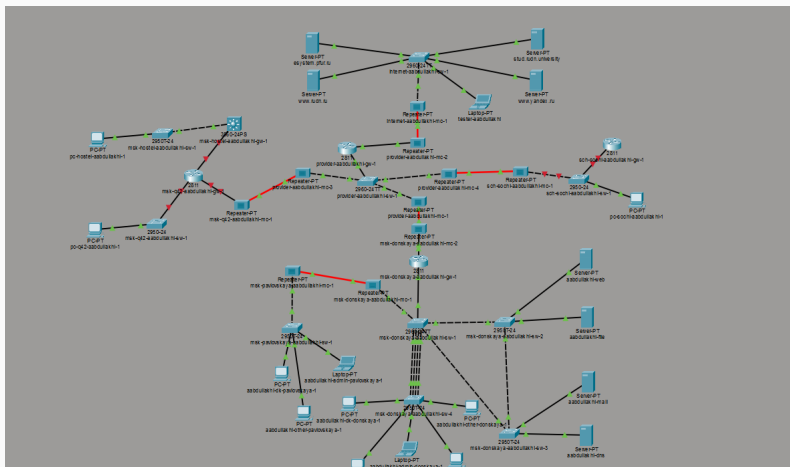


Рис. 1: Обновлённая логическая схема проекта



Рис. 2: Физическая схема связи между площадками

Физическая схема размещения площадок в Москве

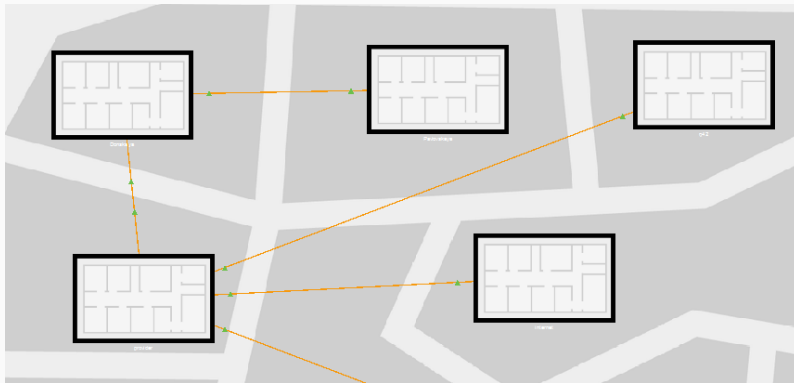


Рис. 3: Физическая схема размещения площадок в Москве

Подсеть основной территории организации 42-го квартала в Москве

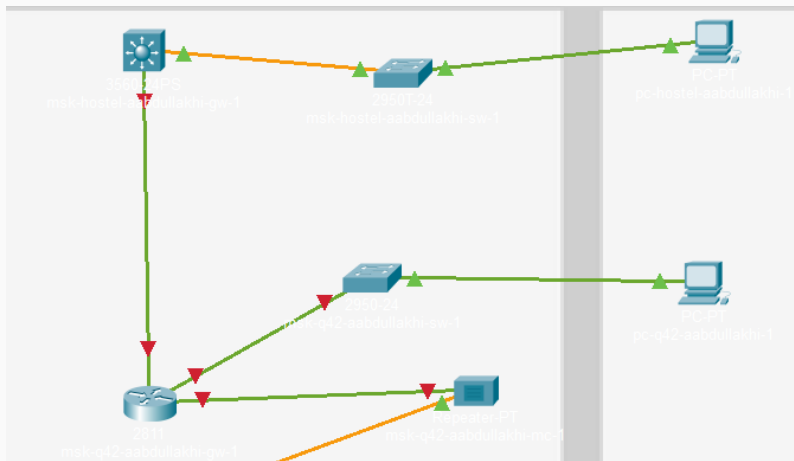


Рис. 4: Подсеть основной территории организации 42-го квартала в Москве

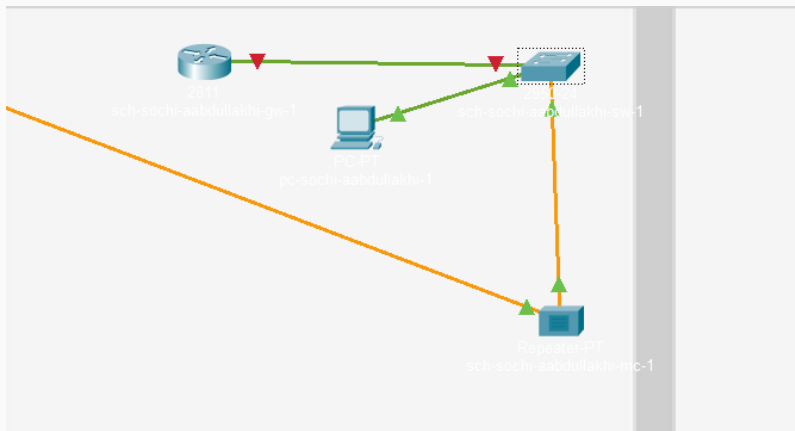
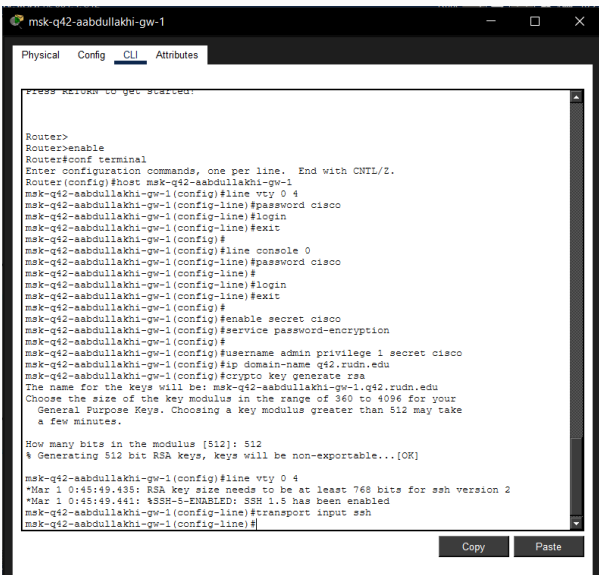


Рис. 5: Подсеть филиала в г. Сочи

Первоначальная настройка маршрутизатора msk-q42-aabdullakhi-gw-1



```
msk-q42-aabdullakhi-gw-1
Physical Config CLI Attributes

Press RETURN to get started:

Router>
Router>enable
Router#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host msk-q42-aabdullakhi-gw-1
msk-q42-aabdullakhi-gw-1(config)#line vty 0 4
msk-q42-aabdullakhi-gw-1(config-line)#password cisco
msk-q42-aabdullakhi-gw-1(config-line)#login
msk-q42-aabdullakhi-gw-1(config-line)#exit
msk-q42-aabdullakhi-gw-1(config)#
msk-q42-aabdullakhi-gw-1(config)#line console 0
msk-q42-aabdullakhi-gw-1(config-line)#password cisco
msk-q42-aabdullakhi-gw-1(config-line)#
msk-q42-aabdullakhi-gw-1(config-line)#login
msk-q42-aabdullakhi-gw-1(config-line)#exit
msk-q42-aabdullakhi-gw-1(config)#
msk-q42-aabdullakhi-gw-1(config)#enable secret cisco
msk-q42-aabdullakhi-gw-1(config)#service password-encryption
msk-q42-aabdullakhi-gw-1(config)#
msk-q42-aabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-q42-aabdullakhi-gw-1(config)#ip domain-name q42.rudn.edu
msk-q42-aabdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-q42-aabdullakhi-gw-1.q42.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-q42-aabdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:45:49.435: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:45:49.441: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-q42-aabdullakhi-gw-1(config-line)#transport input ssh
msk-q42-aabdullakhi-gw-1(config-line)#
```

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Первоначальная настройка коммутатора msk-q42-aabdullakhi-sw-1

```
msk-q42-aabdullakhi-sw-1
Physical Config CLI Attributes

Switch>enable
Switch#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-q42-aabdullakhi-sw-1
msk-q42-aabdullakhi-sw-1(config)#line vty 0 4
msk-q42-aabdullakhi-sw-1(config-line)#password cisco
msk-q42-aabdullakhi-sw-1(config-line)#login
msk-q42-aabdullakhi-sw-1(config-line)#exit
msk-q42-aabdullakhi-sw-1(config)#line console 0
msk-q42-aabdullakhi-sw-1(config-line)#password cisco
msk-q42-aabdullakhi-sw-1(config-line)#login
msk-q42-aabdullakhi-sw-1(config-line)#exit
msk-q42-aabdullakhi-sw-1(config)#
msk-q42-aabdullakhi-sw-1(config)#enable secret cisco
msk-q42-aabdullakhi-sw-1(config)#service password-encryption
msk-q42-aabdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-q42-aabdullakhi-sw-1(config)#
msk-q42-aabdullakhi-sw-1(config)#ip domain-name q42-rudn.rdu
msk-q42-aabdullakhi-sw-1(config)#crypto key generate rsa
% Invalid input detected at '^' marker.

msk-q42-aabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-q42-aabdullakhi-sw-1.q42-rudn.rdu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

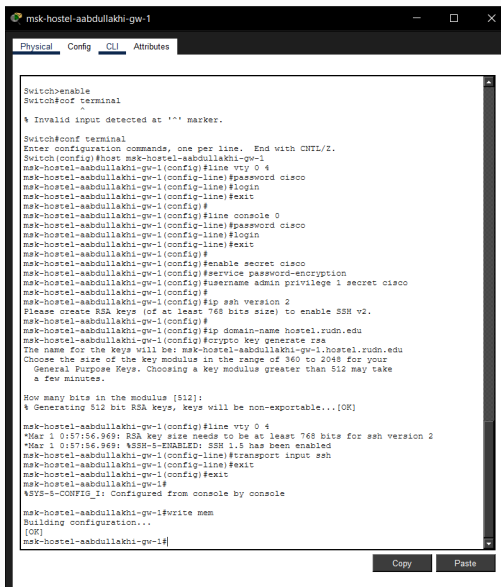
msk-q42-aabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:52:9.565: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:52:9.565: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-q42-aabdullakhi-sw-1(config-line)#transport input ssh
msk-q42-aabdullakhi-sw-1(config-line)#exit
msk-q42-aabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-aabdullakhi-sw-1#write mem
Building configuration...
[OK]
msk-q42-aabdullakhi-sw-1#
```

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Первоначальная настройка маршрутизатора msk-hostel-aabdullakhi-gw-1



```
msk-hostel-aabdullakhi-gw-1
Physical Config CLI Attributes

Switch>enable
Switch#conf terminal
^
% Invalid input detected at '^' marker.

Switch#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-hostel-aabdullakhi-gw-1
msk-hostel-aabdullakhi-gw-1(config)#line vty 0 4
msk-hostel-aabdullakhi-gw-1(config-line)#password cisco
msk-hostel-aabdullakhi-gw-1(config-line)#login
msk-hostel-aabdullakhi-gw-1(config-line)#exit
msk-hostel-aabdullakhi-gw-1(config)#
msk-hostel-aabdullakhi-gw-1(config)#line console 0
msk-hostel-aabdullakhi-gw-1(config-line)#password cisco
msk-hostel-aabdullakhi-gw-1(config-line)#login
msk-hostel-aabdullakhi-gw-1(config-line)#exit
msk-hostel-aabdullakhi-gw-1(config)#
msk-hostel-aabdullakhi-gw-1(config)#enable secret cisco
msk-hostel-aabdullakhi-gw-1(config)#service password-encryption
msk-hostel-aabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-hostel-aabdullakhi-gw-1(config)#
msk-hostel-aabdullakhi-gw-1(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
msk-hostel-aabdullakhi-gw-1(config)#
msk-hostel-aabdullakhi-gw-1(config)#ip domain-name hostel.rudn.edu
msk-hostel-aabdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-hostel-aabdullakhi-gw-1.hostel.rudn.edu
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

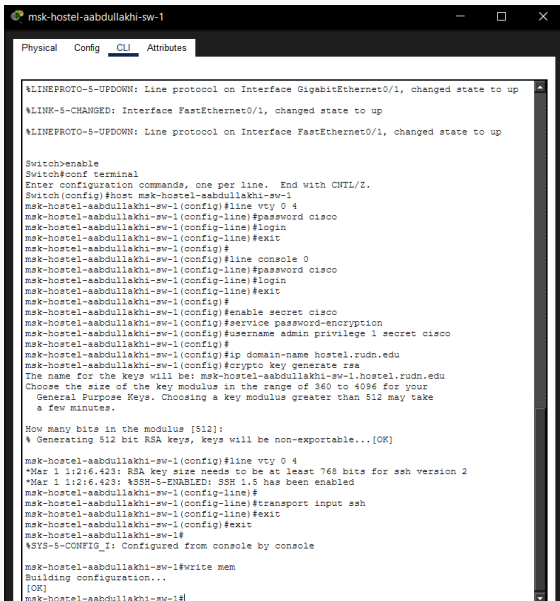
msk-hostel-aabdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:57:56.969: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:57:56.969: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-hostel-aabdullakhi-gw-1(config-line)#transport input ssh
msk-hostel-aabdullakhi-gw-1(config-line)#exit
msk-hostel-aabdullakhi-gw-1(config)#exit
msk-hostel-aabdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-aabdullakhi-gw-1#write mem
Building configuration...
[OK]
msk-hostel-aabdullakhi-gw-1#
```

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Первоначальная настройка коммутатора msk-hostel-aabdullakhi-sw-1



```
msk-hostel-aabdullakhi-sw-1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

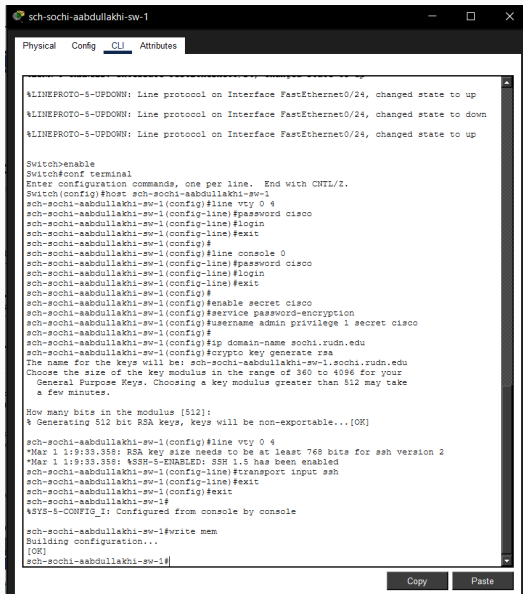
Switch>enable
Switch#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-hostel-aabdullakhi-sw-1
msk-hostel-aabdullakhi-sw-1(config)#line vty 0 4
msk-hostel-aabdullakhi-sw-1(config-line)#password cisco
msk-hostel-aabdullakhi-sw-1(config-line)#login
msk-hostel-aabdullakhi-sw-1(config-line)#exit
msk-hostel-aabdullakhi-sw-1(config)#
msk-hostel-aabdullakhi-sw-1(config)#line console 0
msk-hostel-aabdullakhi-sw-1(config-line)#password cisco
msk-hostel-aabdullakhi-sw-1(config-line)#login
msk-hostel-aabdullakhi-sw-1(config-line)#exit
msk-hostel-aabdullakhi-sw-1(config)#
msk-hostel-aabdullakhi-sw-1(config)#enable secret cisco
msk-hostel-aabdullakhi-sw-1(config)#service password-encryption
msk-hostel-aabdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-hostel-aabdullakhi-sw-1(config)#
msk-hostel-aabdullakhi-sw-1(config)#ip domain-name hostel.rudn.edu
msk-hostel-aabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-hostel-aabdullakhi-sw-1.hostel.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-hostel-aabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 1:2:6.423: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:2:6.423: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-hostel-aabdullakhi-sw-1(config-line)#
msk-hostel-aabdullakhi-sw-1(config-line)#transport input ssh
msk-hostel-aabdullakhi-sw-1(config-line)#exit
msk-hostel-aabdullakhi-sw-1(config)#exit
msk-hostel-aabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-aabdullakhi-sw-1#write mem
Building configuration...
[OK]
msk-hostel-aabdullakhi-sw-1#
```

Первоначальная настройка коммутатора sch-sochi-aabdullakhi-sw-1



```
sch-sochi-aabdullakhi-sw-1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch>enable
Switch#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host sch-sochi-aabdullakhi-sw-1
sch-sochi-aabdullakhi-sw-1(config)#line vty 0 4
sch-sochi-aabdullakhi-sw-1(config-line)#password cisco
sch-sochi-aabdullakhi-sw-1(config-line)#login
sch-sochi-aabdullakhi-sw-1(config-line)#exit
sch-sochi-aabdullakhi-sw-1(config)#
sch-sochi-aabdullakhi-sw-1(config)#line console 0
sch-sochi-aabdullakhi-sw-1(config-line)#password cisco
sch-sochi-aabdullakhi-sw-1(config-line)#login
sch-sochi-aabdullakhi-sw-1(config-line)#exit
sch-sochi-aabdullakhi-sw-1(config)#
sch-sochi-aabdullakhi-sw-1(config)#enable secret cisco
sch-sochi-aabdullakhi-sw-1(config)#service password-encryption
sch-sochi-aabdullakhi-sw-1(config)#username admin privilege 1 secret cisco
sch-sochi-aabdullakhi-sw-1(config)#
sch-sochi-aabdullakhi-sw-1(config)#ip domain-name sochi.rudn.edu
sch-sochi-aabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: sch-sochi-aabdullakhi-sw-1.sochi.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

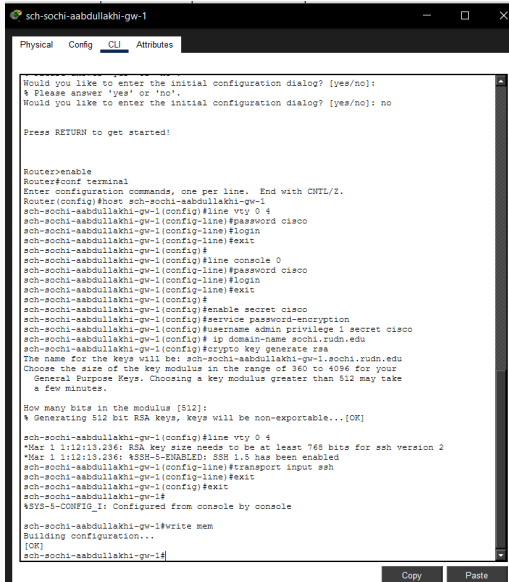
How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

sch-sochi-aabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 1:9:33.358: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:9:33.358: %SSH-5-ENABLED: SSH 1.5 has been enabled
sch-sochi-aabdullakhi-sw-1(config-line)#transport input ssh
sch-sochi-aabdullakhi-sw-1(config-line)#exit
sch-sochi-aabdullakhi-sw-1(config)#exit
sch-sochi-aabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-aabdullakhi-sw-1#write mem
Building configuration...
[OK]
sch-sochi-aabdullakhi-sw-1#
```

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Первоначальная настройка маршрутизатора sch-sochi-aabdullakhi-gw-1



```
sch-sochi-aabdullakhi-gw-1
Physical Config CLI Attributes

Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host sch-sochi-aabdullakhi-gw-1
sch-sochi-aabdullakhi-gw-1(config)#line vty 0 4
sch-sochi-aabdullakhi-gw-1(config-line)#password cisco
sch-sochi-aabdullakhi-gw-1(config-line)#login
sch-sochi-aabdullakhi-gw-1(config-line)#exit
sch-sochi-aabdullakhi-gw-1(config)#
sch-sochi-aabdullakhi-gw-1(config)#line console 0
sch-sochi-aabdullakhi-gw-1(config-line)#password cisco
sch-sochi-aabdullakhi-gw-1(config-line)#login
sch-sochi-aabdullakhi-gw-1(config-line)#exit
sch-sochi-aabdullakhi-gw-1(config)#
sch-sochi-aabdullakhi-gw-1(config)#enable secret cisco
sch-sochi-aabdullakhi-gw-1(config)#service password-encryption
sch-sochi-aabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
sch-sochi-aabdullakhi-gw-1(config)# ip domain-name sochi.rudn.edu
sch-sochi-aabdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: sch-sochi-aabdullakhi-gw-1.sochi.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

sch-sochi-aabdullakhi-gw-1(config)#line vty 0 4
*Mar 1 1:12:13.236: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:12:13.236: %SSH-5-ENABLED: SSH 1.5 has been enabled
sch-sochi-aabdullakhi-gw-1(config-line)#transport input ssh
sch-sochi-aabdullakhi-gw-1(config-line)#exit
sch-sochi-aabdullakhi-gw-1(config)#exit
sch-sochi-aabdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-aabdullakhi-gw-1#write mem
Building configuration...
[OK]
sch-sochi-aabdullakhi-gw-1#
```

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Вывод

В ходе лабораторной работы схема проекта в Cisco Packet Tracer была дополнена: в корпоративную сеть добавлены подсеть основной территории организации 42-го квартала в Москве и подсеть филиала в г. Сочи. Для новых площадок размещены маршрутизаторы, коммутаторы и оконечные устройства, выполнено подключение к провайдерскому сегменту.

Для добавленного оборудования выполнена первоначальная настройка: имена устройств, пароли, шифрование, локальные пользователи, доменные имена, RSA-ключи, SSH. Конфигурации сохранены.